LÂM GIA PHÚC NGUYÊN

COMPUTER SCIENCE STUDENT

Ho Chi Minh City, Vietnam | contact@rawsie.dev | github.com/rawsie-dev

Computer science student with experience in machine learning research, computer vision, software development, and competitive programming. Experienced in designing experiments, developing technical systems, and leading student teams.

EDUCATION

VNU-HCM High School for the Gifted (PTNK) **August 2024 – May 2027 (Expected)**

High School Diploma, Informatics Specialization

- **GPA:** 9.7/10 (Grade 9); 9.6/10 (Grade 10); 9.6/10 (Grade 11)
- **Standardized testing:** SAT: 1520 | IELTS: 8.0 | AP Calculus BC: 5, AP Computer Science A: 5, AP Computer Science Principles: 4.
- **Relevant coursework:** Machine Learning and Deep Learning, and Data Structures & Algorithms.

RESEARCH & ACADEMIC EXPERIENCE

Brain-Inspired Neural Networks **June 2026 – Present**

First author

- Investigated functional diversity and competitive sparsity as brain-inspired approaches to improving artificial neural networks.
- Designed and evaluated parameter-free neural network modifications through controlled experiments on tabular and image classification tasks.
- Evaluated effects on feature redundancy and generalization without introducing additional trainable parameters; authored a first-author research manuscript [View paper].

Synthetic Image Detection Research **June 2025 – October 2025**

Co-author

- Conducted research on lightweight detection of AI-generated images as part of the MediaEval 2025 campaign.
- Built core data-processing and evaluation pipelines, analyzed the dataset, and led experimental runs.
- Contributed to a published study exploring lightweight synthetic-image detection using distilled contrastive representations. **Published** through the MediaEval 2025 platform [View paper].

INTERNSHIP EXPERIENCE

BOCC - Invoice Information Extractor **May 2026 – Present**

Software Engineering Intern

- Identified workflow automation opportunities to streamline manual document processing, architecting a secure offline desktop application for extracting structured information from Vietnamese invoices.
- Integrated PaddleOCR, VietOCR, and optional local LLM inference into a PySide6 desktop application, handling interface development and packaging.
- Enabled local invoice processing without cloud-based inference, supporting data privacy and reducing reliance on manual information entry.

EXTRACURRICULAR EXPERIENCE & LEADERSHIP

The Gifted Battlefield (Student Organization)

A non-profit academic platform providing free and specialized study materials for gifted middle and high school students throughout the country

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Vice Head of Informatics Department **2025 – Present**

- Co-led an 18-member team in organizing mock tests and informatics competitions, supervised preparation and execution of academic activities and events, oversaw the development and review of specialized competitive programming materials, ensuring technical accuracy and quality.
- Developed and reviewed competitive programming materials for regional informatics competitions and mock examinations serving 50+ high school participants; supported event operations for middle and high school students.

Member, Informatics Department **2024 – 2025**

- Contributed to the Informatics Department's academic initiatives.
- Assisted with technical preparation for mock examinations and academic events.
- Supported the delivery of accessible competitive-programming activities for students.

Chùa Vô Ưu teaching

Volunteer, Mathematics Instructor (Grade 8) **June 2025 – August 2025 & June 2026 – August 2026**

- Volunteered to teach mathematics to Grade 8 students during two consecutive summer terms.
- Designed and prepared math lessons, assignments, tests, and exams tailored to students' learning needs and taught two cohorts of eight students.
- Supported 16 student learning experiences through personalized pacing and a supportive classroom environment.

Informatics Major's Traditional Festival

Vice Head of Events Committee **June 2025 – February 2026**

- Helped organize the traditional festival for the Informatics major.
- Pitched festival games and a stamp-and-reward system; helped to write the MC script and an event proposal; staffed a station on the day of the event.

Hoa Phượng Đỏ (Student Organization)

Member, Đội hình bình dân học vụ số **June 2026 – September 2026**

- Participated in community digital-literacy activities supporting access to essential digital services.
- Developed supporting materials, including a presentation on essential Vietnamese apps for older adults, and assisted with in-person activities.
- Supported community members in learning about practical digital tools and services.

PiMA 2026 MATHEMATICS AND APPLICATIONS SUMMER CAMP

Camper **August 2026**

- Attended a summer camp exploring applications of mathematics across science, technology, and AI.
- Participated in lectures and collaborative problem-solving and applied Hidden Markov Models to predict gene function from histone marks to a group research project.

PROJECTS

Personal Website **2026**

Developed a personal website featuring my background, projects and blog posts using SvelteKit and MDsveX [rawsie.dev].

Piano Music Visualizer **2026**

Built a browser-based falling-note visualizer showcasing original piano compositions using SvelteKit, TypeScript, Tone.js and MIDI [piano.rawsie.dev].

ACHIEVEMENTS

- **National Round Qualifier**, Vietnam Olympiad in Artificial Intelligence (VOAI), 2026
  - Qualified for the national stage among the top 246 high school participants nationwide competing for the international delegation.
- **Top 8 (Consolation Prize)**, PIISE 2025, a school-level science and innovation competition; exhibited research at the competition.
- **Consolation Prize**, National *Tin Học Trẻ*, 2024
- **First Prize,** City-level *Tin Học Trẻ* (Youth Informatics Competition), 2020

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, HTML/CSS, LaTeX
- **Frameworks & Libraries:** PyTorch, TensorFlow (Keras), PaddleOCR, VietOCR, PySide6
- **Core Competencies:** Machine Learning, Computer Vision, Data Structures & Algorithms

ADDITIONAL INFORMATION

- **Languages:**
  - Vietnamese (Native)
  - English (Fluent)
- **Interests:**
  - Machine Learning Research
  - Competitive Programming
  - Music (since Grade 5)